

Human-Centred Research Automation: Agentic AI Framework for Literature Discovery, Filtering, and Gap Identification

Ayan Mulla

CeADAR - Ireland's Centre for AI
Dublin, Ireland
ayan.a.mulla9@gmail.com

Yilin Li

CeADAR - Ireland's Centre for AI
Dublin, Ireland
yilin.li@ucd.ie

Donghyeok Lee

CeADAR - Ireland's Centre for AI
Dublin, Ireland
donghyeok.lee@ucd.ie

Alireza Dehghani

CeADAR - Ireland's Centre for AI
Dublin, Ireland
alireza.dehghani@ucd.ie

Abstract

Recent advances in agentic AI raise opportunities to automate complex research tasks; however, the EU AI Act and Human-Centred AI (HCAI) principles emphasise that such automation must safeguard trust, transparency, and responsibility. This work-in-progress study developed an automated framework using agentic AI to support SLRs, which are critical for synthesising knowledge and identifying research gaps but remain resource-intensive, time-consuming, and in need of trust and transparency in their processes. The study presents a human-centred, agentic AI framework for discovering, filtering, analysing academic papers, and identifying research gaps. The proposed framework combines machine efficiency with human oversight to ensure contextual interpretation and responsible judgment. In a pilot study using Llama 3.1 8b, it achieved 71.05% accuracy for abstract filtering compared to a human-led SLR baseline. It also generates structured outputs, such as research gaps, potential directions, and candidate solutions, that enhance clarity and evaluability of the automated synthesis. Preliminary findings demonstrate that automation makes the initial synthesis faster but the human involvement in the later stage ensures that the result can be reliable and transparent. To further assess the framework's alignment with HCAI principles, the next phase will conduct qualitative researcher interviews to evaluate perception of trust and transparency, as well as measuring how structured outputs improve interpretability. Overall, the study contributes to the design of responsible and trustworthy AI systems for research, demonstrating how agentic AI can make SLRs more efficient, reliable, and accessible, while advancing discussions at the intersection of agentic AI, systematic reviews, and human-centred research practice.

CCS Concepts

• **Computing methodologies** → *Multi-agent systems*; Planning for deterministic actions; *Multi-agent planning*.

Keywords

Agentic AI, human-centred AI, systematic literature reviews, trust, transparency

ACM Reference Format:

Ayan Mulla, Donghyeok Lee, Yilin Li, and Alireza Dehghani. 2026. Human-Centred Research Automation: Agentic AI Framework for Literature Discovery, Filtering, and Gap Identification. In *Human Centred Artificial Intelligence - Education and Practice (HCAI-EP '26)*, January 21–22, 2026, Kildare, Ireland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3777490.3777511>

1 Introduction

Systematic literature reviews (SLRs) are important synthesising tools that are used to identify gaps in the existing knowledge, evaluate evidence, and conduct research. Nevertheless, they are both time and resource demanding and they may need several months of study on the part of the researcher [4, 7]. As the large language models (LLMs) and agentic AI systems are becoming more common, researchers are starting to consider ways in which these tools can be used to assist or automate SLR workflows, including the initial discovery and filtering and synthesis. Although these developments will result in high levels of efficiency, there are fears of transparency, accountability and trust amongst the researchers. Total automation in reviews will compromise interpretability and can obscure human accountability of the research result. The EU AI Act and the HCAI frameworks state that AI systems should have human oversight, accountability and responsibility [13]. Our ongoing research suggests a compromise solution, where agentic AI will be efficient, and human experts will be used to make interpretations. The goal is to find out how automation will be used to expedite portions of the review process without compromising transparency and trust.

2 Background

2.1 Agentic AI

Agentic AI can be described as those AI systems which are not merely passively predictive, but instead act autonomously, plan, and perform multi-step tasks. In contrast to traditional machine learning models, agentic systems are able to coordinate tools, access information, and make the use of reason in iterative ways. This causes them to be applicable to complicated workflows like SLRs, in which more than one task (query generation, screening, analysis, synthesis) is to be synchronized. The current advances in agents



This work is licensed under a Creative Commons Attribution 4.0 International License. *HCAI-EP '26, Kildare, Ireland*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2153-3/26/01

<https://doi.org/10.1145/3777490.3777511>

based on LLM show that these systems have a potential to combine retrieval, planning, and reasoning of structured research problems.

Table 1: Applications of Agentic AI across industries and domains.

Domain	Example Applications
Healthcare	Automated clinical trial screening, medical literature summarisation, patient triage support [6]
Education	Intelligent tutoring systems, personalised learning assistants, semi-automated grading [1]
Finance	Fraud detection workflows, compliance monitoring, financial market analysis [1]
Scientific Research	Systematic literature reviews, experiment planning, hypothesis generation [2, 8–10, 14]
Legal	Contract analysis, case law retrieval, due diligence automation [1]
Software Engineering	Automated code review, bug triaging, documentation generation [1]
Business Operations	Market intelligence, customer support agents, supply chain optimisation [1]

2.2 Agentic AI in Research

Agency AI provides a chance to overcome literature review and knowledge synthesis bottlenecks in a research context. In the traditional SLRs, screening of hundreds or thousands of papers is done manually which is cumbersome and prone to error. Agentic AI can alleviate this load by formulating queries, sorting abstracts, summarizing full-text articles and pointing out valuable techniques or results. Earlier systems like ASReview [14] and RobotReviewer [6] prove that semi-automated solutions are viable, whereas more recent systems based on the use of LLM are applied in the context of summarisation, thematic extraction, and gap identification. Nevertheless, the majority of existing applications have problems of reproducibility, interpretability, and trustworthiness.

2.3 Ethical Consideration in Agentic AI

There are ethical issues in using agentic AI in research. To start with, transparency is paramount: the researchers should be aware of how the decisions to include or exclude are made. Second, humans should be held accountable, because AI systems can bring about unseen biases or mistakes. Third, excessive dependence on automation will threaten to remove the element of human judgment, which is essential in research and critique. The regulatory frameworks (e.g., the EU AI Act) point to the fact that AI applications involving high risks have to incorporate the elements of trust, fairness, and responsibility protection. Therefore, the automation frameworks of research must not only feature an increased efficiency but also interpretability and human control.

3 Related Work

Automation in evidence synthesis is over 10-year old. One, like AS-Review, can use active learning to speed up screening of abstracts [14], whereas another tool, RobotReviewer, is an automated tool to estimate risk of bias in clinical trials [6]. Such systems produce efficiency benefits that are measurable, but frequently lack transparency and explainability and so researchers are unable to trust the results completely. More recent multi-agent systems e.g. LatteReview [9] also automate multiple steps of the review pipeline, but still emphasize the role of human supervision of quality and accountability.

LLMs have also been studied as potentially helpful in tasks such as summarisation, classification, and knowledge extraction in review [2, 11]. Indicatively, Oami et al. [8] demonstrated that a biomedical review could have an over 95% decrease in manual work with high specificity when using LLMs. In a survey of close to 3,800 studies, Scherbakov et al. [12] discovered that although the use of LLMs in reviews is becoming more popular, most applications are not full-pipeline but only part of the workflow. Retrieval-augmented generation (RAG) methods have also been suggested to base the results of LLM on evidence, although reproducibility and interpretability issues are still a problem.

LLMs have been found to be subject to biases and hallucinations [1], which is why hybrid and human-based solutions are needed that will allow human beings to control interpretative judgments [13]. A number of recent articles suggest semi-automatic systems in which human verification of outputs is performed by AI [10].

Our work is based on this direction although it is different in two important aspects: we actively incorporate the human-centred AI principles of trust, transparency and accountability in the pipeline and we produce structured outputs, such as research gaps, directions and candidate solutions that extend beyond the efficiency of automated synthesis to enable the interpretability and evaluability of automated synthesis.

4 Human-Centred Research Agentic AI Framework

4.1 Framework Pipeline

The framework has three principles:

- **Human-in-the-loop** - Human beings check and verify AI outputs which makes them accountable.
- **Structured synthesis** - The outputs consist of research gaps, candidate solutions and future directions, pointing towards the clarity and the evaluability.
- **Responsible automation** - The system values transparency and accountability which are in line with HCAI values.

There are four primary stages of the framework:

Discovery: Academic papers were retrieved in databases like Scopus and arXiv.

- *Research Topic Agent (LLM-based):* Modeled around schema database engines like arXiv and produces topic related queries.
- *Paper Search Agent:* Retrieves papers on the basis of APIs and query results.

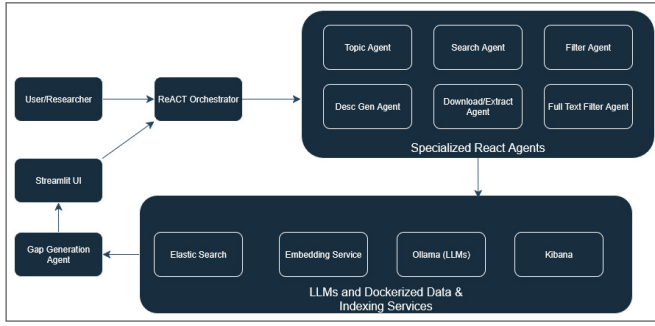


Figure 1: The Human-Centred Agentic AI Framework Pipeline.

Filtering: Llama 3.1 8B is an automated abstract screening system, screened on a human-annotation SLR dataset.

- **Filter Agent (Title + Abstract):** Llama 3.1 is used to determine the relevance or irrelevance of the papers.
- **Description Generation Agent:** It generates more detailed description of the topic of the research to further filter descriptions contextually.
- **Download/Extract Agent:** Retrieves PDFs and decomposes the entire text with PyMuPDF.
- **Full-Text Filter Agent:** The papers are divided into parts (section-based strategy) and rated in terms of relevancy with the help of Llama 3.1.

Analysis: Dissection of themes, methods, and findings of research in the literature.

Gap Identification: Structured outputs (research gaps, potential directions, and candidate solutions) to be reviewed by humans are generated.

- **Gap Generation Agent:** Applies a Retrieval-Augmented Generation (RAG) approach to identify gaps, solutions, and directions.

This modular pipeline ensures efficiency in filtering while maintaining interpretability and enabling human oversight at key checkpoints.

4.2 HCAI Design in Framework

The HCAI considerations are integrated in each of the stages of the pipeline:

- **Transparency:** It logs and audits the outputs at every stage.
- **Responsibility:** Ultimate decisions are still under human consideration.
- **Trust:** Structured outputs (gap keywords, occurrences, comparative analyses) make results verifiable.

5 Methodology

5.1 Dataset

The data were based on the systematic literature review by Lee et al. [5], who examined the aspects of sustainability regarding AI development. Their search query was the following: (green AI) OR (sustainable AI) AND (sustainability) OR (AI pipeline) OR (AI

development) and was performed in Scopus, resulting in 1,094 preliminary studies. The 1,090 that survived after duplicate elimination was subsequently filtered at title/abstract, full-text and eligibility levels according to the PRISMA guidelines. After the elimination of unnecessary items, 150 articles were selected as the final sample. In our work, we reused this data to analyze the automation in systematic reviews, which presents a valuable and quality corpus with breadth (more than 1,000 candidate papers) and topicality (AI sustainability).

5.2 Screening

Several approaches were used to conduct the screening process:

- **Llama 3.1 8B (LLM-based screening):** Used as the primary agentic AI model when it comes to the abstract classification and mirrors the inclusion/exclusion criteria of the initial review.
- **TF-IDF and BM25 (IR baselines):** Models of classical information retrieval as lightweight relevance scorers.
- **Sentence-BERT and SciBERT embeddings:** Dense retrieval and FAISS indexing of semantic similarity matching.

Both methods were given the same set of abstracts and they were required to make relevance predictions. To maintain the principles of Human-Centred AI, all automated predictions were subject to a human-in-the-loop protocol. In this protocol, uncertain or borderline cases flagged by the models (e.g., low confidence thresholds, conflicts between methods, or disagreement with gold-standard criteria) were reviewed manually by domain experts. Human reviewers could override or confirm model predictions, and their feedback was iteratively logged to refine subsequent runs. This ensured that human judgment remained the final arbiter in ambiguous cases, preserving transparency and accountability while still benefiting from automation.

5.3 Screening Evaluation

Evaluation was conducted by comparing the results of automated screening to the gold-standard of human annotations in the works by Lee et al. [5]. Measures were accuracy, precision, recall and F1-score. The time spent processing a single paper was also taken as an indicator of trade-offs between accuracy and efficiency.

Table 2: Performance of different screening methods.

Method	Accuracy (%)
LLM - Llama 3.1	71.05
SentenceBert	68.49
TFIDF Filter	63.83
BM25 Hybrid Filter	62.83
Faiss + SciBert	60.20

Findings revealed that Llama 3.1 8B had the best performance of 71.05% accuracy and was much better than classical IR algorithms (TF-IDF: 63.8%, BM25: 62.8%) and embedding-based algorithms (Sentence-BERT: 68.5%, FAISS+SciBERT: 60.2%). But the slowest (10-30s per abstract) was the LLM approach in contrast to the near-instant retrieval using TF-IDF and BM25.

These results indicate that in spite of quite good alignment that the LLMs offer to human decisions, the traditional approaches have a value to offer in speed sensitive situations. A hybrid between lightweight retrieval to filter and LLM validation may be the best.

5.4 Gap Generation

title	research_gap	gap_direction	gap_solution
Green AI: Exploring Carbon Footprints Mitigation Strategies	This paper addresses the environmental impact of AI systems, particularly in machine learning model training. It tackles the significant challenge of reducing CO2 emissions in large language model development while maintaining performance standards. The work fills a critical gap in understanding the trade-offs between model effectiveness and environmental sustainability in AI research.	Empirical study approach combining performance evaluation with environmental impact assessment	The paper proposes strategies for reducing carbon emissions through optimized training procedures, hardware selection, and model efficiency improvements. It evaluates different approaches using CO2 tracking tools and performance metrics to demonstrate sustainable AI practices.

Figure 2: Structured Research Gap Analysis

In addition to screening, the framework automates the initial synthesis process by identifying candidate research gaps from the included studies. It returns structured outputs that can help researchers speed up the SLR process. A human-in-the-loop mechanism was applied: automatically generated gap statements were not accepted at face value but were validated by researchers, who assessed their plausibility and relevance before integrating them into the final synthesis. This ensured that gap generation served as an assistive tool rather than a replacement for scholarly judgment. This approach builds on automation methods of previous review tools (e.g., RobotReviewer, ASReview) but replaces classification with knowledge generation, allowing one to detect under-researched areas for future investigation while maintaining human responsibility.

Prompt Example for Reproducibility. To ensure methodological transparency, we include below a representative example of the prompt used by the agent responsible for research gap identification:

You are an expert academic research assistant. Your task is
 ↳ to analyze a research paper
 and provide a structured gap analysis.

Input:

- Research Topic: {topic}
- Paper Content: {text_chunk}

Instructions:

1. Identify the RESEARCH_GAPS ADDRESSED: Specify the exact
 ↳ gaps or problems the paper targets.
2. Identify the RESEARCH_DIRECTION: Describe the general
 ↳ approach or methodology of the paper in 20-40 words.
 ↳
3. Identify the SOLUTION_APPROACH: Describe in 60-100 words
 ↳ how the paper addresses the gaps, including methods,
 ↳ algorithms, frameworks, or experiments.
4. Provide the output in the following structured format:

RESEARCH_GAPS: [Detailed description, 120-180 words]

RESEARCH_DIRECTION: [20-40 words]

SOLUTION_APPROACH: [60-100 words]

Rules:

- Do not add commentary outside these fields.
- Use complete sentences and concise wording.
- If information is missing, indicate "Not specified."

This prompt guided the model in generating structured gap analyses for each included study, ensuring consistency, reproducibility, and ease of downstream parsing.

6 Discussion

We have preliminarily found three outcomes linked to each other to demonstrate opportunities and challenges of incorporating agentic AI into systematic literature reviews (SLRs). To begin with, automation is proven to speed up the previous steps of the review pipeline, especially when it comes to the process of abstract filtering, during which the tasks that normally take weeks to complete can be accomplished in hours. This advantage of efficiency is essential to scholars who encounter ever-growing amounts of scholarly literature, but also makes one wonder of the tradeoff between expediency and depth of interpretation. Second, the importance of human oversight is supported by our findings even though it is evident that large language models offer better performance in the screening tasks. Automated systems will not ensure accuracy of context itself and protection against possible biases and errors, particularly in situations when the decisions to include or exclude are unclear. Human intervention will guarantee the maintenance of transparency, trust, and accountability, which are applicable to the guidelines of Human-Centred AI (HCAI) that focus on making responsible judgments and not on the results provided by machines. Third, the systematic outputs produced by our framework, including well phrased research gaps, candidate solutions, and future directions, transcend efficiency benefits and actually improve the interpretation of automated synthesis. Both of these outputs generate information that can be put into practice by the researcher, but it allows the researcher to critically assess the review process itself and create a more transparent and auditable knowledge production pipeline. Collectively, these results point to the possibility of hybrid solutions, in which the effectiveness of agentic AI is offset by the situational accuracy of human scientists. These hybrid systems can constitute a viable direction to the future, as it provides a compromise between the two extremes of full automation and the traditional manual reviews, and at the same time, resolve the issue of reliability, interpretability, and trustworthiness of the outputs by the researcher.

7 Future Work

The next stage will involve:

- Conducting qualitative researcher interviews to assess perceptions of trust, transparency, and accountability.
- Expanding evaluation to measure interpretability improvements from structured outputs.
- Evaluating the results of better LLMs
- Refining gap generation using hybrid RAG + graph databases to improve relational accuracy across papers.
- Scaling experiments to larger datasets and testing newer embedding models (e.g., bge-base, gte-large).

8 Conclusion

This preliminary paper presents a human-centred framework that balances the efficiency of agentic AI with the need for trust, transparency, and human oversight in Systematic Literature Reviews (SLRs). By introducing human control into an otherwise automated literature discovery and synthesis process, the framework implements key principles of Human-Centred AI (HCAI). Initial findings are encouraging: the pilot study achieved 71.05% accuracy in abstract filtering, demonstrating the potential of AI-assisted screening while highlighting the need for human validation. The framework also enhances interpretability and accountability by producing structured and verifiable outputs, such as identified research gaps and candidate solutions. Although the screening performance indicates promise, missing relevant studies or including irrelevant ones could impact synthesis outcomes, particularly in sensitive domains. Human oversight mitigates these risks, ensuring that expert judgment complements automated outputs. Furthermore, as LLMs may inherit biases from training data, structured outputs and transparency logs provide mechanisms to detect and reduce bias propagation. Overall, this work offers a feasible approach for designing responsible AI systems that make SLRs more efficient, reliable, and interpretable, supporting trustworthy and human-centred research practices.

References

- [1] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, et al. 2021. On the opportunities and risks of foundation models. *arXiv:2108.07258* [cs.LG]
- [2] C. Cao, R. Arora, P. Cento, K. Manta, E. Farahani, M. Cecere, et al. 2025. Automation of systematic reviews with large language models. doi:10.1101/2025.06.13.25329541
- [3] X. Fan, K. Truong, et al. 2023. A systematic review of Green AI. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 13, 4 (2023), e1507. doi:10.1002/widm.1507
- [4] Barbara Kitchenham and Stuart Charters. 2007. *Guidelines for performing systematic literature reviews in software engineering*. Technical Report EBSE-2007-01. Keele University and Durham University.
- [5] D. Lee, Y. Li, et al. 2025. Towards sustainable AI development: challenges, opportunities, and the Sustainable AI Development Card. In *Proceedings of the UK Academy for Information Systems Conference 2025 (UKAIS '25)*. <https://aisel.aisnet.org/ukais2025/22/> Paper 22.
- [6] Iain J. Marshall, Joël Kuiper, and Byron C. Wallace. 2015. RobotReviewer: evaluation of a system for automatically assessing bias in clinical trials. *Journal of the American Medical Informatics Association* (2015). doi:10.1093/jamia/ocv044 Published online 2015 July 24.
- [7] David Moher, Alessandro Liberati, Jennifer Tetzlaff, Douglas G. Altman, and The PRISMA Group. 2009. Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLoS Medicine* 6, 7 (2009), e1000097. doi:10.1371/journal.pmed.1000097
- [8] Takuya Oami, Yukinori Okada, Taka-aki Nakada, et al. 2024. Using a large language model to screen citations for a systematic review reduced workload by 95%. *JAMA Network Open* 7, 7 (2024), e2420496. doi:10.1001/jamanetworkopen.2024.20496
- [9] Parnia Rouzrokhi and M. Shariatnia. 2025. LatteReview: a multi-agent framework for systematic review automation using large language models. *arXiv:2501.05468* [cs.CL]
- [10] M. Ruan, J. Fan, M. Liu, Z. Meng, X. Zhang, and C. Zhang. 2025. Artificial intelligence for the science of evidence synthesis: how good are AI-powered tools for automatic literature screening? *BMC Medical Research Methodology* (2025). Version of record available via PMC12376440.
- [11] X. Ruan, J. Smith, et al. 2025. Can large language models match the conclusions of systematic reviews? *arXiv:2505.22787* [cs.CL]
- [12] D. Scherbakov, N. Hubig, V. Jansari, A. Bakumenko, and L. A. Lenert. 2024. The emergence of large language models (LLM) as a tool in literature reviews: an LLM automated systematic review. *Journal of the American Medical Informatics Association* 32, 5 (2024). doi:10.1093/jamia/ocaf063
- [13] Ben Shneiderman. 2020. Human-centered artificial intelligence: reliable, safe & trustworthy. *International Journal of Human-Computer Interaction* 36, 6 (2020), 495–504. doi:10.1080/10447318.2020.1741118
- [14] Rens Van de Schoot, Jonathan de Bruin, R. Schram, S. Winter, G. Maassen, et al. 2020. An open source machine learning framework for efficient and transparent systematic reviews. *Nature Machine Intelligence* 3 (2020), 125–133. doi:10.1038/s42256-020-00287-7 Published 2021.